Chapter 19, Problem 24P

Problem

Find the resistive circuit that represents these *y* parameters:

$$[\mathbf{y}] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{8} \end{bmatrix} \text{S}$$

Step-by-step solution

Step 1 of 2

$$[Y] = \begin{bmatrix} \frac{1}{2} & \frac{-1}{4} \\ \frac{-1}{4} & \frac{3}{8} \end{bmatrix}$$

Since $Y_{12} = Y_{21}$ the two port Net work is reciprocal.

$$Y_{11} - Y_{12} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$Y_{22} - Y_{12} = \frac{3}{8} - \frac{1}{4} = \frac{1}{8}$$

Step 2 of 2